Characterization of novel ADPr antibodies

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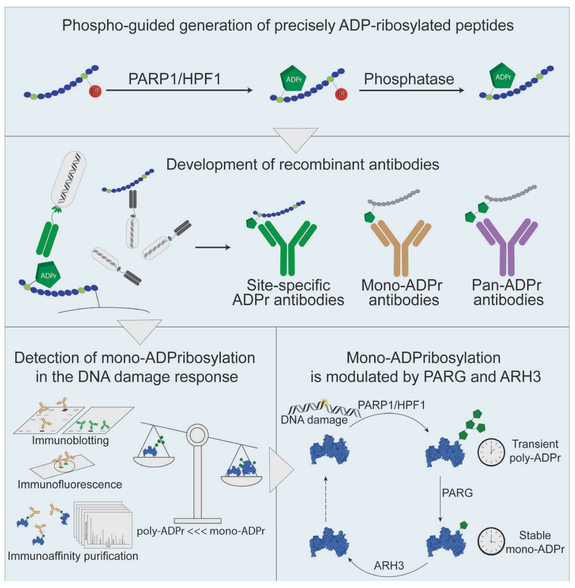
Abstract

- Phospho-guided strategy generates precisely ADPribosylated peptides

- Recombinant site-specific and broad-specificity antibodies to ADPr were developed

- PARP1 and histone H3 are predominantly mono-ADPribosylated upon DNA damage

- Cellular levels of PARP1 and histone H3 mono-ADPr are modulated by PARG and ARH3



Peptide and Antibody generation

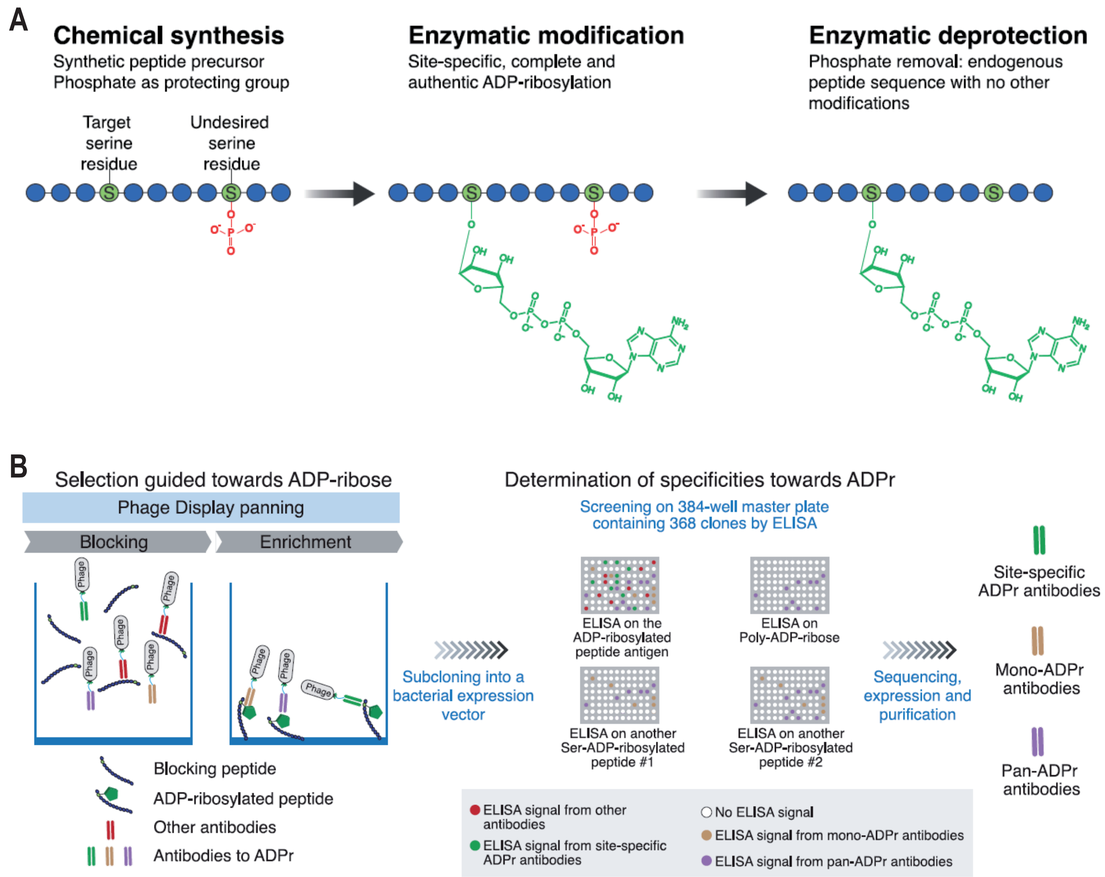


Figure 1. Recombinant, Site-Specific Antibodies to ADPr on PARP1 and Histone H3
(A) Robust, Scalable, and Complete Enzymatic ADPr peptides. A three-step strategy for site-specific ADPr.
(B) Schematic representation of the phage-display screen and selection strategy.

Mono- and Pan ADPr Antibody Validation

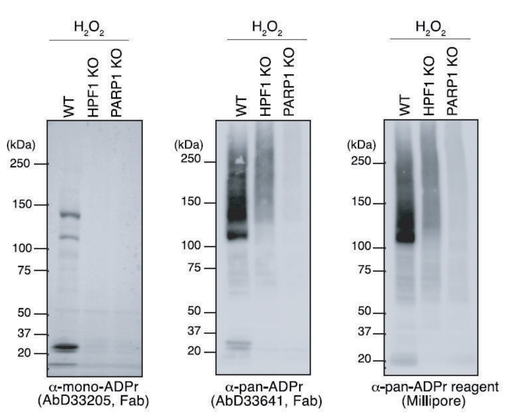


Figure 2. Pan- and Mono-specific Recombinant Antibodies to ADP-Ribosylation
Immunoblot analysis of cell extracts from H2O2-treated wild-type (WT), HPF1 KO, and PARP1 KO cells with the indicated antibodies.

Prevalance of Mono-ADPr upon DNA damage

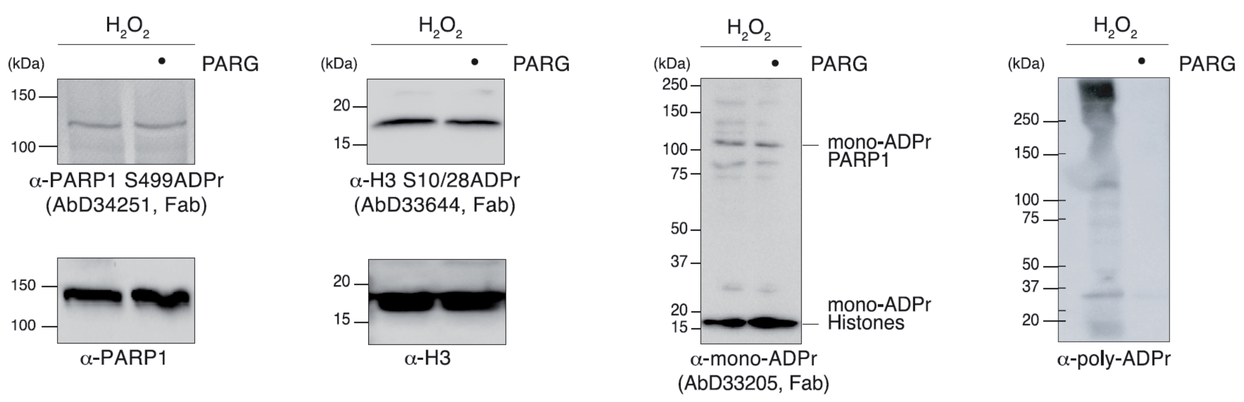


Figure 3. Prevalence of Mono-ADPr upon DNA Damage
Immunoblot analysis of H2O2-treated WT HeLa cell lysatesthat were incubated or not with recombinant PARG.

Prevalance of Mono-ADPr depends on PARG

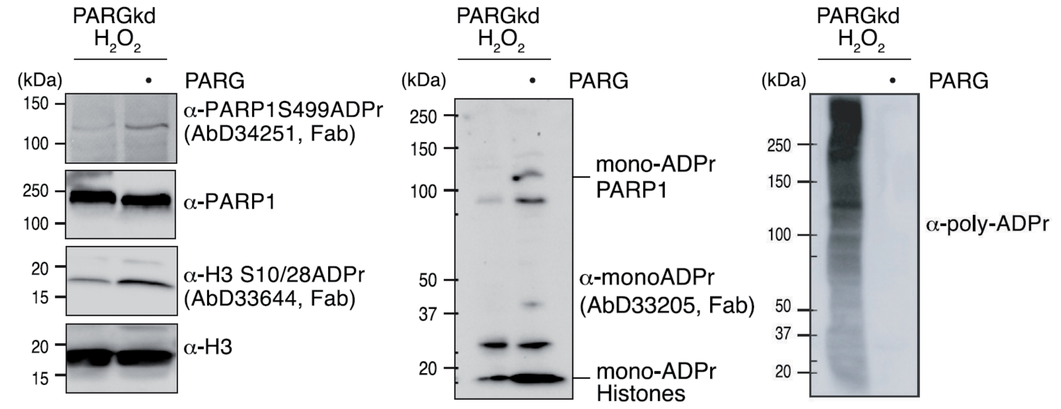


Figure 4. Prevalence of Mono-ADPr Depends on PARG
Immunoblot analysis of H2O2-treated PARG KD cell lysates incubated or not with recombinant PARG.

Dramatic Effects of ARH3 on Cellular Mono-ADPr Dynamics

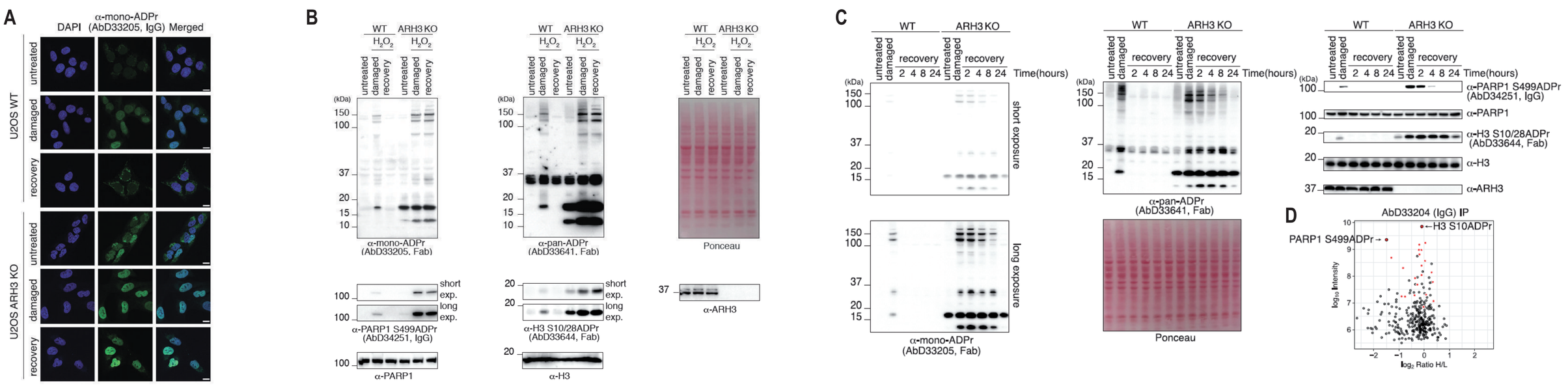


Figure 5. Dramatic Effect of ARH3 on Cellular Mono-ADPr Dynamics
(A) Immunofluorescent staining of WT and ARH3 KO U2OS cells with AbD33205. Scale bar, 16 mM.
(B) Western blot analysis of WT and ARH3 KO U2OS cells, untreated or treated with 2 mM H2O2 for 10 min, followed by a 2-h recovery where indicated.
(C) Immunoblot analysis of WT and ARH3 KO U2OS cells, untreated or treated with 2 mM H2O2. After H2O2 treatment, the cells were switched to drug-free medium for the indicated times.
(D) SILAC quantification of ADPr sites (red dots) comparing ratios from H2O2-treated ARH3 KO (light) cells with those from ARH3 KO cells after 4 h recovery (heavy). Empty dots represent unmodified peptides.

Acknowledgments

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References

1. Bonfiglio J et al. (2020) Cell 183, 1086–1102